INCH-POUND

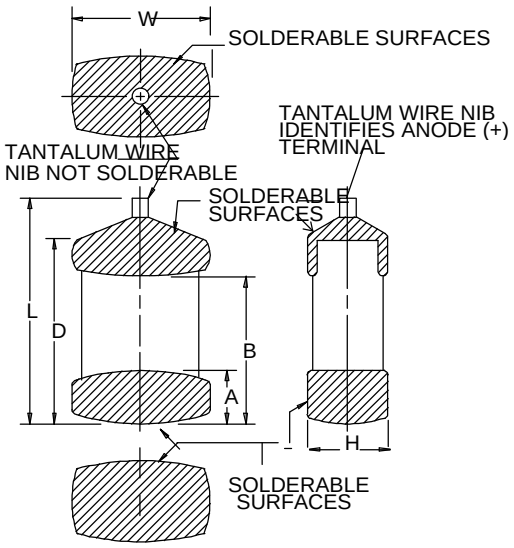
MIL-PRF-55365/7B
3 July 1997
SUPERSEDING
MIL-C-55365/7A
23 August 1990

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLE CWR10

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55365.



Inches	mm	Inches	mm
0.005	0.13	0.069	1.75
0.008	0.20	0.072	1.83
0.010	0.25	0.079	2.01
0.015	0.38	0.085	2.16
0.018	0.46	0.104	2.64
0.020	0.51	0.115	2.92
0.027	0.69	0.136	3.45
0.040	1.02	0.138	3.51
0.045	1.14	0.200	5.08
0.048	1.22	0.235	5.97
0.050	1.27	0.243	6.17
0.051	1.30	0.273	6.93
0.063	1.60	0.275	6.98

Size code	L	W	H	A	B	D
C	.079 ±.008	.045 ±.010	.045 ±.010	.018 ±.008	.050 ±.010	.063 ±.008
R	.273 ±.008	.235 ±.010	.136 ±.010	.040 ±.020	.200 ±.027	.243 ±.010
S	.138 ±.005	.072 ±.008	.048 ±.008	.020 ±.010	.085 ±.015	.115 ±.010
V	.138 ±.005	.104 ±.010	.051 ±.010	.020 ±.010	.085 ±.015	.115 ±.010
X	.275 ±.010	.104 ±.010	.051 ±.010	.040 ±.020	.200 ±.027	.243 ±.010
Y	.275 ±.010	.104 ±.010	.069 ±.010	.040 ±.020	.200 ±.027	.243 ±.010
Z	.275 ±.010	.104 ±.010	.104 ±.010	.040 ±.020	.200 ±.027	.243 ±.010

FIGURE 1. Style CWR10 (conformally coated) capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These capacitors are designed for mounting by dip soldering, vapor phase, reflow soldering, or other conventional means.
4. These capacitors will have a minimum of two bonding pads (top and bottom), on the major (width) dimension.
5. The anode termination (dimension D minus dimension B) shall be .010 inch (0.25 mm) minimum.

FIGURE 1. Style CWR10 (conformally coated) capacitors - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Termination finish: In accordance with MIL-PRF-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365 (see table I).

Resistance to soldering heat: In accordance with MIL-PRF-55365.

Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Shall not exceed tolerance of table I value.
	DF	-	See table I.
	ESR	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.

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Step 4 (+85°C): DCL - See table I.
ΔCap. - Shall not exceed ±10 percent of initial +25°C measurement.
DF - See table I.

Step 5 (+125°C): DCL - See table I.
ΔCap. - Shall not exceed ±15 percent of initial +25°C measurement.
DF - See table I.

Step 6 (+25°C): DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial +25°C measurement.
DF - See table I.
ESR - See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
DF - See table I.
ESR - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

Surge voltage: In accordance with MIL-PRF-55365.

Part or Identifying Number (PIN): See table I. The term Part or Identifying Number (PIN) is equivalent to the term "type designation" previously used in this specification.

TABLE I. Style CWR10 characteristics.

Pin 1/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR10C-335---	4	3.3	0.5	5.0	6.0	6	8	8	6.000	C
CWR10C-685---	4	6.8	1.0	10.0	12.0	6	8	8	2.000	S
CWR10C-106---	4	10.0	1.0	10.0	12.0	6	9	12	2.000	S
CWR10C-156---	4	15.0	1.0	10.0	12.0	6	9	12	1.500	V
CWR10C-226---	4	22.0	1.5	15.0	18.0	6	9	12	1.000	X
CWR10C-336---	4	33.0	3.0	30.0	36.0	6	9	12	1.000	X
CWR10C-476---	4	47.0	3.0	30.0	36.0	6	9	12	0.600	Y
CWR10C-686---	4	68.0	3.0	30.0	36.0	6	9	12	0.600	Y
CWR10C-107---	4	100.0	4.0	40.0	48.0	6	9	12	0.400	Z
CWR10C-337---	4	330.0	13.5	135.0	162.0	10	12	15	0.080	R
CWR10D-225---	6	2.2	0.5	5.0	6.0	6	8	8	7.000	C
CWR10D-475---	6	4.7	1.0	10.0	12.0	6	8	8	2.000	S
CWR10D-685---	6	6.8	1.0	10.0	12.0	6	8	8	2.000	S
CWR10D-106---	6	10.0	1.0	10.0	12.0	6	8	10	1.500	V
CWR10D-156---	6	15.0	1.5	15.0	18.0	6	8	10	1.000	X
CWR10D-226---	6	22.0	3.0	30.0	36.0	6	9	12	1.000	X

See footnote at end of table

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TABLE I. Style CWR10 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR10D-476---	6	47.0	3.0	30.0	36.0	6	9	12	0.600	Y
CWR10D-107---	6	100.0	6.0	60.0	72.0	6	9	12	0.400	Z
CWR10D-187---	6	180.0	13.5	135.0	162.0	10	12	15	0.080	R
CWR10D-227---	6	220.0	13.5	135.0	162.0	10	12	15	0.080	R
CWR10F-155---	10	1.5	0.5	5.0	6.0	6	8	8	10.000	C
CWR10F-335---	10	3.3	1.0	10.0	12.0	6	8	8	2.500	S
CWR10F-475---	10	4.7	1.0	10.0	12.0	6	8	8	2.000	S
CWR10F-685---	10	6.8	1.0	10.0	12.0	6	8	8	2.000	V
CWR10F-106---	10	10.0	1.5	15.0	18.0	6	8	8	1.500	X
CWR10F-156---	10	15.0	3.0	30.0	36.0	6	8	8	1.000	X
CWR10F-226---	10	22.0	3.0	30.0	36.0	6	8	8	0.600	Y
CWR10F-336---	10	33.0	3.0	30.0	36.0	6	8	8	0.400	Z
CWR10F-476---	10	47.0	5.0	50.0	60.0	6	8	8	0.400	Z
CWR10F-107---	10	100.0	10.0	100.0	120.0	10	12	15	0.075	R
CWR10F-127---	10	120.0	12.0	120.0	144.0	10	12	15	0.070	R
CWR10F-127---	10	150.0	15.0	150.0	180.0	10	12	15	0.065	R
CWR10H-105---	15	1.0	0.5	5.0	6.0	6	8	8	10.000	C
CWR10H-225---	15	2.2	1.0	10.0	12.0	6	8	8	3.500	S
CWR10H-335---	15	3.3	1.0	10.0	12.0	6	8	8	2.500	S
CWR10H-475---	15	4.7	1.0	10.0	12.0	6	8	8	2.000	V
CWR10H-685---	15	6.8	1.5	15.0	18.0	6	8	8	1.500	X
CWR10H-106---	15	10.0	1.5	15.0	18.0	6	8	8	1.500	X
CWR10JH156---	15	15.0	2.0	20.0	24.0	6	8	8	0.600	Y
CWR10H-226---	15	22.0	3.0	30.0	36.3	6	8	8	0.400	Z
CWR10H-336---	15	33.0	5.0	50.0	60.0	6	8	8	0.400	Z
CWR10H-686---	15	68.0	10.5	105.0	126.0	6	8	10	0.095	R
CWR10J-684---	20	.68	0.5	5.0	6.0	6	8	8	12.000	C
CWR10J-155---	20	1.5	1.0	10.0	12.0	6	8	8	3.500	S
CWR10J-225---	20	2.2	1.0	10.0	12.0	4	6	8	3.500	S
CWR10J-335---	20	3.3	1.5	15.0	18.0	4	6	8	3.000	V
CWR10J-475---	20	4.7	1.5	15.0	18.0	4	6	8	1.500	X
CWR10J-685---	20	6.8	2.0	20.0	24.0	6	8	8	1.500	X
CWR10J-106---	20	10.0	3.0	30.0	36.0	6	8	8	1.000	Y
CWR10J-156---	20	15.0	3.0	30.0	36.0	6	8	8	0.600	Z
CWR10J-226---	20	22.0	4.5	45.0	54.0	6	8	8	0.400	Z
CWR10J-476---	20	47.0	9.5	95.0	114.0	6	8	8	0.110	R
CWR10K-474---	25	.47	0.5	5.0	6.0	4	6	8	15.000	C
CWR10K-684---	25	.68	0.5	5.0	6.0	4	6	8	5.000	S
CWR10K-105---	25	1.0	1.0	10.0	12.0	4	6	8	3.500	S
CWR10K-155---	25	1.5	1.0	10.0	12.0	6	8	8	3.500	S
CWR10K-225---	25	2.2	1.0	10.0	12.0	4	6	8	2.000	V
CWR10K-335---	25	3.3	1.0	10.0	12.0	6	8	8	1.500	X
CWR10K-475---	25	4.7	12.0	120.0	144.0	6	8	8	1.500	X
CWR10K-685---	25	6.8	3.0	30.0	36.0	6	8	8	1.000	Y

See footnote at end of table

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TABLE I. Style CWR10 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C)	Cap. (nom)	DC Leakage (max)			Dissipation factor (max)			Max ESR 100 kHz +25°C	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
	Volts	µF	µA	µA	µA	%	%	%	Ohms	
CWR10K-106---	25	10.0	2.5	25.0	30.0	6	8	8	1.000	Y
CWR10K-156---	25	15.0	4.0	40.0	48.0	6	8	8	0.600	Z
CWR10K-336---	25	33.0	8.5	85.0	102.0	6	8	8	0.130	R
CWR10M-104---	35	0.1	0.5	5.0	6.0	4	6	8	20.000	S
CWR10M-154---	35	.15	0.5	5.0	6.0	4	6	8	18.000	S
CWR10M-224---	35	.22	0.5	5.0	6.0	4	6	8	15.000	S
CWR10M-334---	35	.33	0.5	5.0	6.0	4	6	8	12.000	S
CWR10M-474---	35	.47	1.0	10.0	12.0	4	6	8	9.000	S
CWR10M-684---	35	.68	1.0	10.0	12.0	4	6	8	5.000	S
CWR10M-105---	35	1.0	1.0	10.0	12.0	4	6	8	3.500	S
CWR10M-155---	35	1.5	1.0	10.0	12.0	4	6	8	3.000	V
CWR10N-225---	35	2.2	2.0	20.0	24.0	6	8	8	2.000	X
CWR10N-335---	35	3.3	2.0	20.0	24.0	4	6	8	1.500	Y
CWR10N-475---	35	4.7	3.0	30.0	36.0	6	8	8	0.800	Z
CWR10N-685---	35	6.8	2.5	25.0	30.0	6	8	8	0.800	Z
CWR10N-106---	35	10.0	3.5	35.0	42.0	6	8	8	0.600	Z
CWR10N-156---	35	15.0	5.5	55.0	66.0	6	8	8	0.190	R
CWR10N-226---	35	22.0	8.0	80.0	96.0	6	8	8	0.160	R

1/ Complete PIN shall include additional symbols to indicate termination finish, capacitance tolerance, and product level.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, CG, MC. OS, SH
Air Force - 17, 19, 99
NASA - NA

Preparing activity:
Army - CR

Agent:
DLA - CC

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